

Kabyadeep Das

Final Year Undergraduate
Majoring in Physics, with a minor in Quantum Technology

Email: kabyadeepdas@iisc.ac.in
Phone: +91 70058 39902

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 – Present	BS (Research)	Indian Institute of Science, Bengaluru	8.1/10
2023	AISSCE (XII)	Kendriya Vidyalaya ONGC, Agartala	95%
2021	AISSE (X)	Kendriya Vidyalaya ONGC, Agartala	95.2%

Scholastic Achievements

- Secured **All India Rank 341** in **IAT 2023** (IISER Aptitude Test) among all applicants nationwide.
- Cleared multiple national-level entrance examinations — **JEE (Mains & Advanced)**, **NEET**, **CUET**, and **NDA** — in the same cycle.
- Recipient of a monthly **UG Institute Scholarship** awarded by the Indian Institute of Science for academic merit.

Work Experience

- Undergraduate Researcher, Prof. Arindam Ghosh’s Group, IISc Bengaluru** (May 2026 – Present)
 - Working on graphene-based nanoscale devices involving Raman spectroscopy, PMMA-mediated wet transfer, and device characterization on SiO₂/Si and a-Si:H substrates.
 - Investigating low-dimensional transport phenomena in graphene devices fabricated via PMMA wet transfer on SiO₂/Si and a-Si:H substrates.
- Undergraduate Research Intern, OUSUMS Lab, IISc Bengaluru** (May 2025 – Jul. 2025)
 - Performed **COMSOL Multiphysics** simulations related to Physical Vapor Deposition of CrCl₃.
 - Assisted in synthesis and characterization of CrCl₃, contributing to the study of 2D magnetic material systems.
 - Developed proficiency in lab instrumentation and data analysis for 2D van der Waals material systems.

Key Projects

- Hartree-Fock Implementation for Small Molecules** (2026)
 - Implemented self-consistent field (SCF) iterations and electronic energy minimization from first principles.
 - Developed RHF code supporting STO-2G through STO-6G basis sets.
 - Computed ground-state energies and validated against reference values;
 - Code hosted at github.com/Kabyakun/Hartree_Fock.
- DFT Calculations using Quantum ESPRESSO** (2026)
 - Used Quantum ESPRESSO to compute electronic band structures, density of states, and convergence behaviour for semiconductor materials.
 - Investigated dependence of calculated properties on plane-wave cutoff and k-point mesh density.
- COMSOL Simulations** (2025)
 - Modelled heat transfer and precursor concentration gradients in a CVD chamber to study CrCl₃ deposition dynamics.
 - Constructed a geometrically accurate model of the CVD setup in COMSOL for simulation validation.
- Infinite OS – Browser Extension** (2024)
 - Built a canvas-style new-tab dashboard with draggable and resizable widgets including clock, timer, sticky notes, todos, YouTube embeds, and quick links.
 - Navigated Mozilla’s extension signing process to get the extension permanently installed in Firefox.

Technical Skills

- Programming Languages:** Python, C, L^AT_EX
- Software and Libraries:** COMSOL Multiphysics, Quantum ESPRESSO, NumPy, Matplotlib, SciPy, PyTorch, Git
- Domain Skills:** Mathematical Modeling, Computational Physics, Quantum Mechanics, Machine Learning
- Experimental Techniques:** Mechanical Exfoliation, CVD, Raman Spectroscopy, Optical Lithography, E-Beam Evaporation, Plasma Asher

Positions of Responsibility

- **Core Coordinator, Rhapsody 4.0** (2026)
 - Headed a team of 11 coordinators and volunteers responsible for managing and liaising with all judges across events at Rhapsody 4.0, IISc’s annual cultural fest.
- **Event Coordinator, Pravega** (2025 – 2026)
 - Coordinated **Shutterbugs**, the nationwide photography competition held as part of IISc’s annual science and cultural festival Pravega.
- **Convenor, IISc Shutter Club** (Jan 2026 – Present)
 - Led the photography and videography club at IISc, overseeing club operations, directing photoshoots, and managing a team for IISc’s student publications and events.
- **Volunteer, IISc Notebook Drive** (2023 – 2024)
 - Participated in educational outreach and support for underprivileged students in government schools.
 - Conducted surveys to identify and shortlist students from disadvantaged backgrounds for equitable scholarship distribution.

Relevant Courses

Classical Mechanics Statistical Mechanics Solid State Physics (Concepts in Solid State Physics) Semiconductor Physics Introduction to Materials for Quantum Technologies Introduction to Quantum Measurement & Control Probability and Statistics	Quantum Mechanics I & II Mathematical Methods of Physics Physics at the Nanoscale 2D Materials Computational Modeling of Materials Introduction to Quantum Computation Machine Learning
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